

4th.py - C:/Users/thala/Downloads/FODS/4th.py (3.14.3)

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```
import numpy as np
```

```
sales_data = np.genfromtxt("Sales_data.csv",  
                           delimiter=";",  
                           skip_header=1,  
                           dtype=str)
```

```
months = sales_data[:,1]  
sales = sales_data[:,4].astype(float)
```

```
Q1 = 0.0  
Q4 = 0.0
```

```
for i in range(len(months)):  
    if ("January" in months[i]) or ("February" in months[i]) or ("March" in months[i]):  
        Q1 += sales[i]  
    elif ("October" in months[i]) or ("November" in months[i]) or ("December" in months[i]):  
        Q4 += sales[i]
```

```
total_sales_year = np.sum(sales)
```

```
percentage_increase = ((Q4 - Q1) / Q1) * 100
```

```
print("Total sales for the year:", round(total_sales_year,2))
```

```
print("Percentage increase from Q1 to Q4:", round(percentage_increase,2), "%")
```

IDLE Shell 3.14.3*

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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

```
>>> ----- RESTART: C:/Users/thala/Downloads/FODS/4th.py -----
```

```
Total sales for the year: 438000.0
```

```
>>> Percentage increase from Q1 to Q4: 74.7 %
```

```
>>> ----- RESTART: C:/Users/thala/Downloads/FODS/4th.py -----
```

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STH.py - C:/Users/thala/Downloads/FODS/STH.py (3.14.3)

File Edit Format Run Options Window Help

import numpy as np

```
fuel_data = np.genfromtxt("Fuel_data.csv",  
                           delimiter=";",  
                           skip_header=1,  
                           dtype=str)
```

```
fuel_efficiency = fuel_data[:,7].astype(float)  
make = fuel_data[:,8]
```

```
average_efficiency = np.mean(fuel_efficiency)
```

```
mazda_eff = fuel_efficiency[make == "mazda"]  
audi_eff = fuel_efficiency[make == "audi"]
```

```
mazda_avg = np.mean(mazda_eff)  
audi_avg = np.mean(audi_eff)
```

```
percentage_improvement = (mazda_avg - audi_avg) / audi_avg * 100
```

```
print("Average Fuel Efficiency of all cars:", round(average_efficiency,2), "MPG")  
print("Average Fuel Efficiency of Mazda:", round(mazda_avg,2), "MPG")  
print("Average Fuel Efficiency of Audi:", round(audi_avg,2), "MPG")  
print("Percentage Improvement from Mazda to Audi:", round(percentage_improvement,2), "%")
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

```
===== RESTART: C:/Users/thala/Downloads/FODS/4th.py =====  
Total sales for the year: 438000.0  
Percentage increase from Q1 to Q4: 74.7 %
```

>>>

```
===== RESTART: C:/Users/thala/Downloads/FODS/4th.py =====  
Total sales for the year: 438000.0  
Percentage increase from Q1 to Q4: 74.7 %
```

>>>

```
===== RESTART: C:/Users/thala/Downloads/FODS/STH.py =====  
Average Fuel Efficiency of all cars: 25.1 MPG  
Average Fuel Efficiency of Mazda: 28.67 MPG  
Average Fuel Efficiency of Audi: 22.0 MPG  
Percentage Improvement from Mazda to Audi: 30.3 %
```

>>>

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```

6th.py - C:/Users/thala/Downloads/FODS/6th.py (3.14.3)
File Edit Format Run Options Window Help
import pandas as pd

df = pd.read_csv("grocerystore.csv")

total_sales = df["Sales"].sum()

print("Total sales for the grocery store:", round(total_sales,2))

discount = 0.10
tax = 0.18

price_after_discount = total_sales - (total_sales * discount)

price_after_tax = price_after_discount + (price_after_discount * tax)

print("Final amount after applying discount and tax:", round(price_after_tax,2))

```

```

IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/4th.py =====
Total sales for the year: 438000.0
Percentage increase from Q1 to Q4: 74.7 %
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/4th.py =====
Total sales for the year: 438000.0
Percentage increase from Q1 to Q4: 74.7 %
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/5th.py =====
Average Fuel Efficiency of all cars: 25.1 MPG
Average Fuel Efficiency of Mazda: 28.67 MPG
Average Fuel Efficiency of Audi: 22.0 MPG
Percentage Improvement from Mazda to Audi: 30.3 %
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/6th.py =====
Total sales for the grocery store: 17100
Final amount after applying discount and tax: 18160.2
>>>

```

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